

Lista IV

Metode Numerice

2021

Funcțiile Beta și Gama (euleriene)

$$\Gamma(p) = \int_0^\infty x^{p-1} e^{-x} dx$$

$$B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx$$

1. $\Gamma(1) = 1$, $\Gamma(p+1) = p\Gamma(p)$, $\Gamma(n) = (n-1)!$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

2. $B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$

Operatorul Bernstein

Notatie $e_k(x) = x^k$

Fie $f : [0, 1] \rightarrow \mathbb{R}$, operatorul B_n definit prin

$$B_n(f)(x) = \sum_{k=0}^n p_{n,k}(x) f\left(\frac{k}{n}\right),$$

unde

$$p_{n,k}(x) = C_n^k x^k (1-x)^{n-k}$$

se numește *operatorul Bernstein*, iar polinomul $B_n(f)$ se numește *polinomul Bernstein*.

1. B_n este un operator liniar ($B_n(\alpha f + \beta g) = \alpha B_n(f) + \beta B_n(g)$)
2. B_n este un operator pozitiv ($f(x) \geq 0 \Rightarrow B_n(f)(x) \geq 0$, $\forall x \in [0, 1]$)
3.
 - (a) $B_n(e_0)(x) = e_0(x) = 1$
 - (b) $B_n(e_1)(x) = e_1(x) = x$
 - (c) $B_n(e_2)(x) = e_2(x) = x^2 + \frac{x(1-x)}{n} = \frac{n-1}{n} e_2(x) + \frac{1}{n} e_0(x)$
 - (d) Dacă $f \in C[0, 1]$ atunci $B_n(f) \Rightarrow f$ (convergență uniformă) pe $[0, 1]$ când $n \rightarrow \infty$

Teorema Bohman-Popoviciu-Korovkin

Fie $L_m : C[a, b] \rightarrow C[a, b]$ un șir de operatori liniari și pozitivi care verifică

$$L_m(e_0)(x) = 1 + u_m(x)$$

$$L_m(e_1)(x) = x + v_m(x)$$

$$L_m(e_2)(x) = x^2 + w_m(x)$$

și $\lim_{m \rightarrow \infty} u_m(x) = \lim_{m \rightarrow \infty} v_m(x) = \lim_{m \rightarrow \infty} w_m(x) = 0$ uniform pe $[a, b]$. Atunci pentru orice $f \in C[a, b]$ avem

$$\lim_{m \rightarrow \infty} L_m(f)(x) = f(x)$$

uniform pe $[a, b]$.

1. Să se determine coeficienții și nodurile formulelor de cuadratură cu grad maxim de exactitate de forma:

$$\begin{aligned}
 \text{(a)} \quad & \int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = Af(-1) + Bf(x_0) + Cf(1) + R(f) \\
 \text{(b)} \quad & \int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = A_0f(x_0) + A_1f(x_1) + A_2f(x_2) + R(f) \\
 \text{(c)} \quad & \int_a^b f(x) dx = Af(a) + Bf\left(\frac{a+b}{2}\right) + Cf(b) + R(f) \\
 \text{(d)} \quad & \int_0^\infty f(x) dx = A_0f(2) + A_1f(x_1) + R(f) \\
 \text{(e)} \quad & \int_1^\infty e^{-2x} f(x) dx = A_0f(x_0) + A_1f(x_1) + R(f) \\
 \text{(f)} \quad & \int_{-\infty}^\infty e^{-x^2} f(x) dx = A_0f(x_0) + A_1f(x_1) + A_2f(x_2) + R(f)
 \end{aligned}$$

Rezolvare:

- (a) În acest caz $n = 2$ (3 noduri) și avem 4 necunoscute x_0 , A , B și C . Vom avea 4 ecuații, $R(e_k) = 0$, $k = 0, 1, 2, 3$, gradul maxim de exactitate este cel puțin 3. Pentru început considerăm $f(x) = (x - x_0)(x^2 - 1)$ și obținem

$$\begin{aligned}
 \int_{-1}^1 \frac{(x^2 - 1)(x - x_0)}{\sqrt{1 - x^2}} dx &= 0A + 0B + 0C \Leftrightarrow \int_{-1}^1 \frac{x^3 - x - x_0(x^2 - 1)}{\sqrt{1 - x^2}} dx = 0 \\
 \Leftrightarrow \underbrace{\int_{-1}^1 \frac{x^3 - x}{\sqrt{1 - x^2}} dx}_{=0} - x_0 \int_{-1}^1 \frac{x^2 - 1}{\sqrt{1 - x^2}} dx &= 0 \Leftrightarrow x_0 \underbrace{\int_{-1}^1 \sqrt{1 - x^2} dx}_{\neq 0} = 0 \Rightarrow x_0 = 0
 \end{aligned}$$

Pentru a determina coeficienții A , B și C considerăm pe rând $f(x) = l_0(x) = (x - 0)(x - 1) = x^2 - x$, $f(x) = l_1(x) = (x + 1)(x - 1) = x^2 - 1$ și $f(x) = l_2(x) = (x + 1)(x - 0) = x^2 + x$. Ajungem la

$$\begin{aligned}
 \bullet \quad & \int_{-1}^1 \frac{x^2 - x}{\sqrt{1 - x^2}} dx = 2A \Leftrightarrow \frac{\pi}{2} = 2A \Leftrightarrow A = \frac{\pi}{4} \\
 \bullet \quad & \int_{-1}^1 \frac{x^2 - 1}{\sqrt{1 - x^2}} dx = -B \Leftrightarrow -\frac{\pi}{2} = -B \Leftrightarrow B = \frac{\pi}{2} \\
 \bullet \quad & \int_{-1}^1 \frac{x^2 + x}{\sqrt{1 - x^2}} dx = 2C \Leftrightarrow \frac{\pi}{2} = 2C \Leftrightarrow C = \frac{\pi}{4}
 \end{aligned}$$

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1 - x^2}} dx = \frac{\pi}{4} (f(-1) + 2f(0) + f(1)) + R(f)$$

- (b) În acest caz $n = 2$ funcția pondere este $w(x) = \frac{1}{\sqrt{1-x^2}}$. Gradul maxim de exactitate va fi $2n + 1 = 5$, iar nodurile vor fi rădăcinile polinomului Cebîșev de speța întâi de grad 3

$$T_3(x) = \cos(3\arccos(x)) = 4x^3 - 3x$$

Așadar $x_0 = -\frac{\sqrt{3}}{2}$, $x_1 = 0$, $x_2 = \frac{\sqrt{3}}{2}$. Considerând pe rând $f(x) = 1$, $f(x) = x$ și $f(x) = x^2$ ajungem la

$$\begin{cases} \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx = A_0 + A_1 + A_2 \\ \int_{-1}^1 \frac{x}{\sqrt{1-x^2}} dx = -\frac{\sqrt{3}}{2} A_0 + 0A_1 + \frac{\sqrt{3}}{2} A_2 \\ \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx = \frac{3}{4} A_0 + 0A_1 + \frac{3}{4} A_2 \end{cases} \Leftrightarrow \begin{cases} A_0 + A_1 + A_2 = \pi \\ \frac{\sqrt{3}}{2} (A_2 - A_0) = 0 \\ \frac{3}{4} (A_0 + A_2) = \frac{\pi}{2} \end{cases} \Leftrightarrow \begin{cases} A_0 = \frac{\pi}{3} \\ A_1 = \frac{\pi}{3} \\ A_2 = \frac{\pi}{3} \end{cases}$$

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = \frac{\pi}{3} \left(f\left(-\frac{\sqrt{3}}{2}\right) + f(0) + f\left(\frac{\sqrt{3}}{2}\right) \right) + R(f)$$

- (c) În acest caz avem $n = 2$, funcția pondere este $w(x) = 1$ și avem 3 necunoscute A, B și C . Pentru a determina aceste necunoscute vom avea nevoie de 3 ecuații și gradul de exactitate va fi cel puțin 2, *i.e.* $R(e_k) = 0, k = 0, 1, 2$. Vom considera pe rând $f(x) = l_a(x) = (x - \frac{a+b}{2})(x - b)$, $f(x) = l_{\frac{a+b}{2}}(x) = (x - a)(x - b)$ și $f(x) = l_b(x) = (x - a)(x - \frac{a+b}{2})$. Ajungem la

$$\begin{aligned} \bullet \int_a^b \left(x - \frac{a+b}{2}\right)(x-b) dx &= Al_a(a) \Leftrightarrow \frac{(b-a)^3}{12} = A \frac{(b-a)^2}{2} \Leftrightarrow A = \frac{b-a}{6} \\ \bullet \int_a^b (x-a)(x-b) dx &= Bl_{\frac{a+b}{2}}\left(\frac{a+b}{2}\right) \Leftrightarrow -\frac{(b-a)^3}{6} = -B \frac{(b-a)^2}{4} \Leftrightarrow B = \frac{2(b-a)}{3} \\ \bullet \int_a^b (x-a)\left(x - \frac{a+b}{2}\right) dx &= Cl_b(b) \Leftrightarrow \frac{(b-a)^3}{12} = C \frac{(b-a)^2}{2} \Leftrightarrow C = \frac{b-a}{6} \\ \int_a^b f(x) dx &= \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f\left(\frac{a+b}{2}\right) + \frac{b-a}{6} f(b) + R(f) \end{aligned}$$

- (d) În acest caz avem $n = 1$ și necunoscutele sunt A_0, A_1, x_1 . Pentru a determina aceste necunoscute avem nevoie de 3 condiții, *i.e.* $R(e_k) = R(x^k) = 0, k = 0, 1, 2$. Considerăm $f(x) = (x-2)(x-x_1)$ și ajungem la

$$\begin{aligned} \int_0^\infty e^{-x}(x-2)(x-x_1) dx &= 0A_0 + 0A_1 \\ \Leftrightarrow \int_0^\infty e^{-x}x^2 dx - (2+x_1) \int_0^\infty e^{-x}x dx + 2x_1 \int_0^\infty e^{-x} dx &= 0 \\ \Leftrightarrow \Gamma(3) - (2+x_1)\Gamma(2) + 2x_1\Gamma(1) &= 0 \Leftrightarrow 2 - (2+x_1) + 2x_1 = 0 \Leftrightarrow x_1 = 0 \end{aligned}$$

Pentru a determina A_0 și A_1 considerăm $f(x) = x$ respectiv $f(x) = x - 2$

$$\begin{aligned} \int_0^\infty e^{-x}x dx &= 2A_0 + 0A_1 \Leftrightarrow 2A_0 = \Gamma(2) \Rightarrow A_0 = \frac{1}{2} \\ \int_0^\infty e^{-x}(x-2) dx &= 0A_0 + (0-2)A_1 \Leftrightarrow -2A_1 = \Gamma(2) - 2\Gamma(1) \Rightarrow A_1 = \frac{1}{2} \end{aligned}$$

$$\int_0^\infty e^{-x}f(x) dx = \frac{1}{2} \left(f(0) + f(2) \right) + R(f)$$

- (e) Facem schimbarea de variabilă $t = 2x - 2$ ($dt = 2dx, x = 1 \Rightarrow t = 0$) și obținem

$$\int_1^\infty e^{-2x}f(x) dx = \frac{1}{2} \int_0^\infty e^{-t-2}f\left(\frac{t+2}{2}\right) dt \quad \underbrace{\quad}_{g(t)=f\left(\frac{t+2}{2}\right)} = \frac{1}{2e^2} \int_0^\infty e^{-t}g(t) dt$$

Problema se reduce la a determina coeficienții și nodurile pentru

$$\int_0^\infty e^{-t}g(t) dt = A'_0g(t_0) + A'_1g(t_1) + R(g)$$

Avem $n = 2$ și funcția pondere $w(x) = e^{-x}$. Gradul maxim de exactitate va fi $2n + 1 = 5$, iar nodurile vor fi rădăcinile polinomului Laguerre de grad 2

$$L_2(t) = ke^t \frac{d^2}{dt^2}(t^2 e^{-t}) = k(t^2 - 4t + 2)$$

Așadar $t_0 = 2 - \sqrt{2}, t_1 = 2 + \sqrt{2}$. Pentru a determina coeficienții A'_0 și A'_1 considerăm pe rând $g(t) = t - (2 + \sqrt{2})$ și $g(t) = t - (2 - \sqrt{2})$. Ajungem la

$$\begin{aligned} \bullet \int_0^\infty e^{-t}(t - (2 + \sqrt{2})) dt &= A'_0(2 - \sqrt{2} - (2 + \sqrt{2})) \Leftrightarrow \Gamma(2) - (2 + \sqrt{2})\Gamma(1) = (-2\sqrt{2})A'_0 \\ \bullet \int_0^\infty e^{-t}(t - (2 - \sqrt{2})) dt &= A'_1(2 + \sqrt{2} - (2 - \sqrt{2})) \Leftrightarrow \Gamma(2) - (2 - \sqrt{2})\Gamma(1) = 2\sqrt{2}A'_1 \end{aligned}$$

$$A'_0 = \frac{2 + \sqrt{2}}{4}, \quad A'_1 = \frac{2 - \sqrt{2}}{4}$$

Ținând cont de transformările efectuate obținem

$$x_0 = \frac{t_0 + 2}{2} = \frac{2 - \sqrt{2} + 2}{2} = \frac{4 - \sqrt{2}}{2}, \quad x_1 = \frac{t_1 + 2}{2} = \frac{2 + \sqrt{2} + 2}{2} = \frac{4 + \sqrt{2}}{2}$$

și

$$A_0 = \frac{1}{2e^2} A'_0 = \frac{2 + \sqrt{2}}{8e^2}, \quad A_1 = \frac{1}{2e^2} A'_1 = \frac{2 - \sqrt{2}}{8e^2}$$

În final

$$\int_1^\infty e^{-2x} f(x) dx = \frac{2 + \sqrt{2}}{8e^2} f\left(\frac{4 - \sqrt{2}}{2}\right) + \frac{2 - \sqrt{2}}{8e^2} f\left(\frac{4 + \sqrt{2}}{2}\right) + R(f)$$

- (f) În acest caz $n = 2$ funcția pondere este $w(x) = e^{-x^2}$. Gradul maxim de exactitate va fi $2n + 1 = 5$, iar nodurile vor fi rădăcinile polinomului Hermite de grad 3

$$H_3(x) = ke^{x^2} \frac{d^3}{dx^3} (e^{-x^2}) = -k(8x^3 - 12x)$$

Așadar $x_0 = -\sqrt{\frac{3}{2}}$, $x_1 = 0$, $x_2 = \sqrt{\frac{3}{2}}$. Pentru a determina coeficienții A_0 , A_1 și A_2 considerăm pe rând $f(x) = l_0(x) = (x - 0)(x - \sqrt{\frac{3}{2}})$, $f(x) = l_1(x) = (x + \sqrt{\frac{3}{2}})(x - \sqrt{\frac{3}{2}})$ și $f(x) = l_2(x) = (x - 0)(x + \sqrt{\frac{3}{2}})$. Ajungem la

$$\begin{aligned} & \bullet \int_{-\infty}^{\infty} e^{-x^2} \left(x^2 - \sqrt{\frac{3}{2}}x\right) dx = A_0 \left(-\sqrt{\frac{3}{2}} - 0\right) \left(-\sqrt{\frac{3}{2}} - \sqrt{\frac{3}{2}}\right) \Leftrightarrow \\ & 3A_0 = \int_{-\infty}^{\infty} e^{-x^2} x^2 dx - \underbrace{\int_{-\infty}^{\infty} e^{-x^2} \sqrt{\frac{3}{2}}x dx}_{=0(\text{impară})} \Leftrightarrow 3A_0 = \int_{-\infty}^{\infty} e^{-x^2} x^2 dx = 2 \int_0^{\infty} e^{-x^2} x^2 dx \stackrel{x^2=t}{=} \\ & = 2 \int_0^{\infty} e^{-t} t \frac{1}{2\sqrt{t}} dt = \int_0^{\infty} e^{-t} t^{\frac{1}{2}} dt = \Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2} \Rightarrow A_0 = \frac{\sqrt{\pi}}{6} \\ & \bullet \int_{-\infty}^{\infty} e^{-x^2} \left(x^2 - \frac{3}{2}\right) dx = A_1 \left(0 - \sqrt{\frac{3}{2}}\right) \left(0 + \sqrt{\frac{3}{2}}\right) \Leftrightarrow \\ & -\frac{3}{2} A_1 = 2 \underbrace{\int_0^{\infty} e^{-x^2} x^2 dx}_{=\frac{\sqrt{\pi}}{2}} - \frac{3}{2} \underbrace{\int_{-\infty}^{\infty} e^{-x^2} dx}_{=\sqrt{\pi}} = -\sqrt{\pi} \Rightarrow A_1 = \frac{2\sqrt{\pi}}{3} \\ & \bullet \int_{-\infty}^{\infty} e^{-x^2} \left(x^2 + \sqrt{\frac{3}{2}}x\right) dx = A_2 \left(\sqrt{\frac{3}{2}} - 0\right) \left(\sqrt{\frac{3}{2}} + \sqrt{\frac{3}{2}}\right) \Leftrightarrow \\ & 3A_2 = \int_{-\infty}^{\infty} e^{-x^2} x^2 dx + \underbrace{\int_{-\infty}^{\infty} e^{-x^2} \sqrt{\frac{3}{2}}x dx}_{=0(\text{impară})} \Leftrightarrow 3A_2 = \frac{\sqrt{\pi}}{2} \Rightarrow A_2 = \frac{\sqrt{\pi}}{6} \\ & \int_{-\infty}^{\infty} e^{-x^2} f(x) dx = \frac{\sqrt{\pi}}{6} f\left(-\sqrt{\frac{3}{2}}\right) + \frac{2\sqrt{\pi}}{3} f(0) + \frac{\sqrt{\pi}}{6} f\left(-\sqrt{\frac{3}{2}}\right) \end{aligned}$$

2. Determinați coeficienții a, b, c, d astfel încât formula de cuadratură

$$\int_{-1}^1 f(x) dx = af(-1) + bf(1) + cf'(-1) + df'(1) + R(f)$$

să aibă gradul de exactitate 3.

Rezolvare: Din faptul că formula are gradul de exactitate 3 deducem că $R(e_0) = R(e_1) = R(e_2) = R(e_3) = 0$ și obținem sistemul

$$\begin{aligned} a + b &= \int_{-1}^1 1 \, dx = 2 \\ a * (-1) + b * 1 + c + d &= \int_{-1}^1 x \, dx = 0 \\ a * (-1)^2 + b * 1^2 + c * 2 * (-1) + d * 2 * 1 &= \int_{-1}^1 x^2 \, dx = \frac{2}{3} \\ a * (-1)^3 + b * 1^3 + c * 3 * (-1)^2 + d * 3 * 1^2 &= \int_{-1}^1 x^3 \, dx = 0 \end{aligned}$$

adică

$$\begin{cases} a + b = 2 \\ -a + b + c + d = 0 \\ a + b - 2c + 2d = \frac{2}{3} \\ -a + b + 3c + 3d = 0 \end{cases}$$

Obținem $a = b = 1$, $c = \frac{1}{3}$, $d = -\frac{1}{3}$ și formula de cuadratură

$$\int_{-1}^1 f(x) \, dx = f(-1) + f(1) + \frac{1}{3}f'(-1) - \frac{1}{3}f'(1) + R(f)$$

3. Să se arate că șirurile de operatori de mai jos sunt șiruri de aproximare

(a) $(M_n)_{n \in \mathbb{N}}$, $M_n : C[0, 1] \rightarrow C[0, 1]$ definiți prin

$$M_n(f)(x) = (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t f(t) p_{n,k}(t) \, dt}{k+1}$$

unde $p_{n,k}(t)$ sunt polinoamele fundamentale Bernstein

(b) $(K_n)_{n \in \mathbb{N}}$, $K_n : C[0, 1] \rightarrow C[0, 1]$ definiți prin

$$K_n(f)(x) = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) \, dt$$

(c) $(U_n)_{n \in \mathbb{N}}$, $U_n : C[0, 1] \rightarrow C[0, 1]$ definiți prin

$$U_n(f)(x) = \frac{1}{3}B_n(f)(x) + \frac{2}{3}(n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) \, dt$$

Rezolvare: Vom aplica teorema Bohman-Popoviciu-Korovkin

(a)

$$\begin{aligned}
M_n(\alpha f + \beta g)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t(\alpha f + \beta g)(t) p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \left(\frac{\int_0^1 t\alpha f(t) p_{n,k}(t) dt}{k+1} + \frac{\int_0^1 t\beta g(t) p_{n,k}(t) dt}{k+1} \right) = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \left(\frac{\alpha \int_0^1 t f(t) p_{n,k}(t) dt}{k+1} + \frac{\beta \int_0^1 t g(t) p_{n,k}(t) dt}{k+1} \right) = \\
&= \alpha(n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t f(t) p_{n,k}(t) dt}{k+1} + \\
&\quad + \beta(n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t g(t) p_{n,k}(t) dt}{k+1} = \\
&= \alpha M_n(f)(x) + \beta M_n(g)(x), \quad \forall f, g \in C[0, 1], \quad \alpha, \beta \in \mathbb{R}, \quad x \in [0, 1].
\end{aligned}$$

Fie $f \in C[0, 1]$, $f \geq 0$, i.e. $f(x) \geq 0$, $\forall x \in [0, 1]$. Avem $xf(x) \geq 0$, $\forall x \in [0, 1] \Rightarrow \int_0^1 tf(t) dt \geq 0$. Ținând cont de faptul că

$$p_{n,k}(x) = C_n^k x^k (1-x)^{n-k} \geq 0, \quad \forall x \in [0, 1]$$

deducem

$$M_n(f)(x) = \underbrace{(n+1)(n+2)}_{\geq 0} \sum_{k=0}^n \underbrace{p_{n,k}(x)}_{\geq 0} \underbrace{\frac{\int_0^1 t f(t) p_{n,k}(t) dt}{k+1}}_{\geq 0}$$

În continuare vom calcula $M_n(e_0)$, $M_n(e_1)$, $M_n(e_2)$.

$$\begin{aligned}
M_n(e_0)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t e_0(t) p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 C_n^k t^{k+1} (1-t)^{n-k} dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\frac{n!}{(n-k)!k!} B(k+2, n-k+1)}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{n!}{(n-k)!(k+1)!} \frac{\Gamma(k+2)\Gamma(n-k+1)}{\Gamma(n+3)} = \\
&= \sum_{k=0}^n p_{n,k}(x) \frac{(n+2)!}{(n-k)!(k+1)!} \frac{(k+1)!(n-k)!}{(n+2)!} = \sum_{k=0}^n p_{n,k}(x) = \\
&= \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} = (x+1-x)^n = 1 = e_0(x)
\end{aligned}$$

$$\begin{aligned}
M_n(e_1)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t e_1(t) p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t^2 p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 C_n^k t^{k+2} (1-t)^{n-k} dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\frac{n!}{(n-k)!k!} B(k+3, n-k+1)}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{n!}{(n-k)!(k+1)!} \frac{\Gamma(k+3)\Gamma(n-k+1)}{\Gamma(n+4)} = \\
&= \sum_{k=0}^n p_{n,k}(x) \frac{(n+2)!}{(n-k)!(k+1)!} \frac{(k+2)!(n-k)!}{(n+3)!} = \sum_{k=0}^n p_{n,k}(x) \frac{k+2}{n+3} = \\
&= \frac{1}{n+3} \sum_{k=0}^n p_{n,k}(x)(k+2) = \frac{n}{n+3} \left(\sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \frac{2}{n} \sum_{k=0}^n p_{n,k}(x) \right) = \\
&= \frac{n}{n+3} (B_n e_1)(x) + \frac{2}{n+3} (B_n e_0)(x) = \frac{n}{n+3} \cdot x + \frac{2}{n+3} \cdot 1
\end{aligned}$$

$$\begin{aligned}
M_n(e_2)(x) &= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t e_2(t) p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 t^3 p_{n,k}(t) dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\int_0^1 C_n^k t^{k+3} (1-t)^{n-k} dt}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{\frac{n!}{(n-k)!k!} B(k+4, n-k+1)}{k+1} = \\
&= (n+1)(n+2) \sum_{k=0}^n p_{n,k}(x) \frac{n!}{(n-k)!(k+1)!} \frac{\Gamma(k+4)\Gamma(n-k+1)}{\Gamma(n+5)} = \\
&= \sum_{k=0}^n p_{n,k}(x) \frac{(n+2)!}{(n-k)!(k+1)!} \frac{(k+3)!(n-k)!}{(n+4)!} = \sum_{k=0}^n p_{n,k}(x) \frac{(k+3)(k+4)}{(n+3)(n+4)} = \\
&= \frac{1}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) (k^2 + 5k + 6) = \frac{n^2}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) \frac{k^2}{n^2} + \\
&+ \frac{5n}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \frac{6}{(n+3)(n+4)} \sum_{k=0}^n p_{n,k}(x) = \\
&= \frac{n^2}{(n+3)(n+4)} (B_n e_2)(x) + \frac{5n}{(n+3)(n+4)} (B_n e_1)(x) + \frac{6}{(n+3)(n+4)} (B_n e_0)(x) = \\
&= \frac{n^2}{(n+3)(n+4)} \left(x^2 + \frac{x(1-x)}{n} \right) + \frac{5n}{(n+3)(n+4)} x + \frac{6}{(n+3)(n+4)}
\end{aligned}$$

Obținem

$$M_n(e_0) = e_0$$

$$M_n(e_1) = e_1 + \left(-\frac{3}{n+3} e_1 + \frac{2}{n+3} e_0 \right)$$

$$M_n(e_2) = e_2 + \left(-\frac{8n+12}{(n+3)(n+4)} e_2 + \frac{6n}{(n+3)(n+4)} e_1 + \frac{6}{(n+3)(n+4)} e_0 \right)$$

(b)

$$\begin{aligned}
K_n(\alpha f + \beta g)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} (\alpha f + \beta g)(t) dt = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \left(\alpha \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt + \beta \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} g(t) dt \right) = \\
&= \alpha(n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt + \beta(n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} g(t) dt = \\
&= \alpha(K_n f)(x) + \beta(K_n g)(x)
\end{aligned}$$

Fie $f \in C[0, 1]$, $f \geq 0$, i.e. $f(x) \geq 0$, $\forall x \in [0, 1]$. Avem

$$K_n(f)(x) = (n+1) \sum_{k=0}^n \underbrace{p_{n,k}(x)}_{\geq 0} \underbrace{\int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} f(t) dt}_{\geq 0} \geq 0$$

În continuare vom calcula $K_n(e_0)$, $K_n(e_1)$, $K_n(e_2)$.

$$\begin{aligned}
K_n(e_0)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} e_0(t) dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} 1 dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \cdot t \Big|_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \frac{1}{n+1} = \sum_{k=0}^n p_{n,k}(x) = (B_n e_0)(x) = e_0(x) = 1 \\
\\
K_n(e_1)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} e_1(t) dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} t dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \cdot \frac{t^2}{2} \Big|_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \frac{1}{2} \left(\frac{(k+1)^2}{(n+1)^2} - \frac{k^2}{(n+1)^2} \right) = \frac{n+1}{2} \sum_{k=0}^n p_{n,k}(x) \frac{2k+1}{(n+1)^2} = \frac{n}{n+1} \sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \\
&+ \frac{1}{2(n+1)} \sum_{k=0}^n p_{n,k}(x) = \frac{n}{n+1} (B_n e_1)(x) + \frac{1}{2(n+1)} (B_n e_0)(x) = \frac{n}{n+1} x + \frac{1}{2(n+1)} \\
\\
K_n(e_2)(x) &= (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} e_2(t) dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \int_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} t^2 dt = (n+1) \sum_{k=0}^n p_{n,k}(x) \cdot \frac{t^3}{3} \Big|_{\frac{k}{n+1}}^{\frac{k+1}{n+1}} = \\
&= (n+1) \sum_{k=0}^n p_{n,k}(x) \frac{1}{3} \left(\frac{(k+1)^3}{(n+1)^3} - \frac{k^3}{(n+1)^3} \right) = \frac{n+1}{3} \sum_{k=0}^n p_{n,k}(x) \frac{3k^2+2k+1}{(n+1)^3} = \\
&= \frac{1}{3(n+1)^2} \sum_{k=0}^n p_{n,k}(x) (3k^2+2k+1) = \frac{n^2}{(n+1)^2} \sum_{k=0}^n p_{n,k}(x) \frac{k^2}{n^2} + \frac{n}{(n+1)^2} \sum_{k=0}^n p_{n,k}(x) \frac{k}{n} + \\
&+ \frac{1}{3(n+1)^2} \sum_{k=0}^n p_{n,k}(x) = \frac{n^2}{(n+1)^2} B_n(e_2)(x) + \frac{n}{(n+1)^2} B_n(e_1)(x) + \frac{1}{3(n+1)^2} B_n(e_0)(x) = \\
&= \frac{n^2}{(n+1)^2} \left(x^2 + \frac{x(1-x)}{n} \right) + \frac{n}{(n+1)^2} x + \frac{1}{3(n+1)^2}
\end{aligned}$$

(c) Observăm că pentru orice $f \in C[0, 1]$ și $x \in [0, 1]$ avem

$$U_n(f)(x) = \frac{1}{3} B_n(f)(x) + \frac{2}{3} K_n(f)(x)$$

Din liniaritatea și pozitivitatea lui B_n și K_n deducem că U_n este un operator liniar și pozitiv. În continuare vom calcula $U_n(e_0)$, $U_n(e_1)$, $U_n(e_2)$.

$$\begin{aligned}
U_n(e_0) &= \frac{1}{3} B_n(e_0) + \frac{2}{3} K_n(e_0) = \frac{1}{3} e_0 + \frac{2}{3} e_0 = e_0 \\
\\
U_n(e_1) &= \frac{1}{3} B_n(e_1) + \frac{2}{3} K_n(e_1) = \frac{1}{3} B_n(e_1) + \frac{2}{3} \left(\frac{n}{n+1} B_n(e_1) + \frac{1}{2(n+1)} B_n(e_0) \right) = \\
&= \frac{3n+1}{3(n+1)} B_n(e_1) + \frac{1}{3(n+1)} B_n(e_0) = \frac{3n+1}{3(n+1)} e_1 + \frac{1}{3(n+1)} e_0 \\
\\
U_n(e_2) &= \frac{1}{3} B_n(e_2) + \frac{2}{3} K_n(e_2) = \frac{1}{3} B_n(e_2) + \frac{2}{3} \left(\frac{n^2}{(n+1)^2} B_n(e_2) + \frac{n}{(n+1)^2} B_n(e_1) + \frac{1}{3(n+1)^2} B_n(e_0) \right) = \\
&= \frac{3n^2+2n+1}{3(n+1)^2} B_n(e_2) + \frac{2n}{3(n+1)^2} B_n(e_1) + \frac{2}{9(n+1)^2} B_n(e_0) = \frac{3n^2+2n+1}{3(n+1)^2} \frac{(n-1)e_2+e_1}{n} + \\
&+ \frac{2n}{3(n+1)^2} e_1 + \frac{2}{9(n+1)^2} e_0 = \frac{3n^3-n^2-n-1}{3n(n+1)} e_2 + \frac{5n^2+2n+1}{3n(n+1)^2} e_1 + \frac{2}{9(n+1)^2} e_0
\end{aligned}$$

4. Fie $p_{n,k}(x)$ polinoamele fundamentale Bernstein. Să se demonstreze următoarea relație de recurență

$$p_{n,k}(x) = (1-x)p_{n-1,k}(x) + xp_{n-1,k-1}(x)$$

Rezolvare: Știm că $p_{n,k}(x) = C_n^k x^k (1-x)^{n-k}$. Avem

$$\begin{aligned} (1-x)p_{n-1,k}(x) + xp_{n-1,k-1}(x) &= \\ &= (1-x)C_{n-1}^k x^k (1-x)^{n-1-k} + xC_{n-1}^{k-1} x^{k-1} (1-x)^{n-1-(k-1)} = \\ &= C_{n-1}^k x^k (1-x)^{n-k} + C_{n-1}^{k-1} x^k (1-x)^{n-k} = x^k (1-x)^{n-k} (C_{n-1}^k + C_{n-1}^{k-1}) = \\ &= C_n^k x^k (1-x)^{n-k} = p_{n,k}(x) \end{aligned}$$

5. Să se calculeze $B_n(e_3)(x)$, unde $B_n : C[0,1] \rightarrow C[0,1]$ sunt operatorii Bernstein, iar $e_k(x) = x^k$

Rezolvare:

$$B_n(f)(x) = \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} f\left(\frac{k}{n}\right)$$

Pentru $e_0(x) = 1$ avem

$$B_n(e_0)(x) = \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} = (x + (1-x))^n = 1$$

Considerăm identitatea

$$(p+q)^n = \sum_{k=0}^n C_n^k p^k q^{n-k}$$

și derivând în funcție de p obținem

$$\frac{d}{dp}((p+q)^n) = n(p+q)^{n-1} = \sum_{k=0}^n C_n^k k p^{k-1} q^{n-k}.$$

Pasul următor constă în a înmulți cu $\frac{p}{n}$ și deducem că

$$\sum_{k=0}^n C_n^k p^k q^{n-k} \left(\frac{k}{n}\right) = p(p+q)^{n-1}.$$

Înlocuind $p = x$ și $q = 1-x$ ajungem la

$$B_n(e_1)(x) = \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} \left(\frac{k}{n}\right) = x(x+1-x)^{n-1} = x$$

Derivând expresia de mai sus de două ori în raport cu p și înmulțind de fiecare dată cu $\frac{p}{n}$ obținem

$$\sum_{k=0}^n C_n^k p^k q^{n-k} \left(\frac{k}{n}\right)^2 = \frac{(n-1)(p+q)^{n-2}}{n} p^2 + \frac{(p+q)^{n-1}}{n} p$$

și

$$\sum_{k=0}^n C_n^k p^k q^{n-k} \left(\frac{k}{n}\right)^3 = \frac{(n-1)(n-2)(p+q)^{n-3}}{n^2} p^3 + \frac{3(n-1)(p+q)^{n-2}}{n^2} p^2 + \frac{(p+q)^{n-1}}{n^2} p$$

Luând $p = x$ și $q = 1 - x$ deducem că

$$B_n(e_2)(x) = \frac{(n-1)x^2}{n} + \frac{x}{n}$$

și

$$B_n(e_3)(x) = \frac{(n-1)(n-2)x^3}{n^2} + \frac{3(n-1)x^2}{n^2} + \frac{x}{n^2}$$

6. Să se calculeze $B_n(f)(x)$ unde $f(x) = \cos(x)$

Rezolvare: Avem

$$B_n(\cos)(x) = \sum_{k=0}^n p_{n,k}(x) \cos\left(\frac{k}{n}\right) = \sum_{k=0}^n x^k (1-x)^{n-k} \cos\left(\frac{k}{n}\right)$$

Considerăm sumele

$$S_1 = \sum_{k=0}^n x^k (1-x)^{n-k} \cos \frac{k}{n}$$

$$S_2 = \sum_{k=0}^n x^k (1-x)^{n-k} \sin \frac{k}{n}$$

Deducem că $B_n(\cos)(x)$ va fi partea reală a sumei $S_1 + iS_2$

$$\begin{aligned} S_1 + iS_2 &= \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} \left(\cos \frac{k}{n} + i \sin \frac{k}{n} \right) = \\ &= \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} \left(\cos \frac{1}{n} + i \sin \frac{1}{n} \right)^k = \\ &= \sum_{k=0}^n C_n^k \left(x \left(\cos \frac{1}{n} + i \sin \frac{1}{n} \right) \right)^k (1-x)^{n-k} = \\ &= \left(x \left(\cos \frac{1}{n} + i \sin \frac{1}{n} \right) + 1 - x \right)^n = \left(x \cos \frac{1}{n} + 1 - x + ix \sin \frac{1}{n} \right)^n \end{aligned}$$

adică

$$\begin{aligned} B_n(\cos)(x) &= C_n^0 \left(x \cos \frac{1}{n} + 1 - x \right)^n + (-1) C_n^2 \left(x \cos \frac{1}{n} + 1 - x \right)^{n-2} \left(x \sin \frac{1}{n} \right)^2 \\ &+ (-1)^2 C_n^4 \left(x \cos \frac{1}{n} + 1 - x \right)^{n-4} \left(x \sin \frac{1}{n} \right)^4 + \dots = \\ &= \sum_{k=0}^n (-1)^k C_n^{2k} \left(x \cos \frac{1}{n} + 1 - x \right)^{n-2k} \left(x \sin \frac{1}{n} \right)^{2k} \end{aligned}$$

Rezolvare II: Folosim formula $\cos x = \frac{e^{ix} + e^{-ix}}{2}$ și liniaritatea operatorului Bernstein B_n și obținem

$$\begin{aligned} B_n(\cos x)(x) &= \frac{1}{2} B_n(e^{ix})(x) + \frac{1}{2} B_n(e^{-ix})(x) = \frac{1}{2} \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} e^{i \frac{k}{n}} + \\ &+ \frac{1}{2} \sum_{k=0}^n C_n^k x^k (1-x)^{n-k} e^{-i \frac{k}{n}} = \frac{1}{2} \sum_{k=0}^n C_n^k \left(x e^{i \frac{1}{n}} \right)^k (1-x)^{n-k} + \frac{1}{2} \sum_{k=0}^n C_n^k \left(x e^{-i \frac{1}{n}} \right)^k (1-x)^{n-k} = \\ &= \frac{1}{2} \left(x e^{i \frac{1}{n}} + 1 - x \right)^n + \frac{1}{2} \left(x e^{-i \frac{1}{n}} + 1 - x \right)^n = \operatorname{Re} \left(x e^{i \frac{1}{n}} + 1 - x \right)^n \end{aligned}$$